The current issue and full text archive of this journal is available at
www.emeraldinsight.com/0957-4093.htm

The development of a framework for route selection in multimodal transportation

Athakorn Kengpol and Sopida Tuammee
Advanced Industrial Engineering Management Systems Research Center, Department of Industrial Engineering, King Mongkut's University of Technology North Bangkok, Bangkok, Thailand, and
Markku Tuominen
Faculty of Industrial Engineering and Management, Lappeenranta University of Technology, Lappeenranta, Finland

Received 29 May 2013
Revised 17 October 2013
Accepted 2 January 2014

Abstract

Purpose – The purpose of this paper is to develop a framework for route selection in multimodal transportation which can reduce cost, lead time, risk and CO_2 emission in multimodal transportation systems.

Design/methodology/approach – This research proposes the development of a framework for route selection in multimodal transportation that includes a six-phase framework to select an optimal multimodal transportation route. The first phase is to collect the data of each route and select the origin and destination. The second phase is to calculate time and cost of each route by using a multimodal transport cost-model. In the third phase, the CO_2 emissions are calculated based upon the 2006 guidelines of Intergovernmental Panel on Climate Change. The fourth phase proposes an integrated quantitative risk assessment, analytic hierarchy process (AHP) and data envelopment analysis methodology to evaluate the multimodal transportation risk. The fifth phase is to prioritize criteria by using the AHP which can be used in the objective function. The final phase is to calculate the optimal route by using the zero-one goal programming.

Findings – The aims of the model are to minimize transportation costs, transportation time, risk and CO_2 emission.

Practical implications – The approach has been tested on a realistic multimodal transportation service, originating from Bangkok in Thailand to a destination at Da Nang port in Vietnam. The results have shown that the approach can provide guidance in choosing the lowest cost route in accordance with other criteria, and to minimize the CO_2 emission effectively.

Originality/value – The contribution of this research lies in the development of a new decision support approach that is flexible and applicable to logistics service providers, in selecting multimodal transportation route under the multi-criteria in term of cost, time, risk and importantly the environmental impact.

Keywords Analytic hierarchy process (AHP), Data envelopment analysis (DEA), Multimodal transport, Multimodal transportation risk, Quantitative risk assessment (QRA), Zero-one goal programming model

Paper type Research paper

1. Introduction

Multimodal transport, as defined by the European Conference of Ministers of Transport, is the combination of two or more modes of transport to move passengers

The authors are very grateful to The Royal Golden Jubilee PhD Program, Thailand Research Fund (TRF) for funding support for this research No. PHD/0002/2551 (Code: 1.I.KN/51/A.1). Thanks go to logistics service providers in Thailand and Vietnam for inquiries about information of transportation route and experts for risk assessment of each route. The authors are very appreciative to all of the anonymous reviewers for their valuable comments.

The International Journal of Logistics Management
Vol. 25 No. 3, 2014
pp. 581-610
© Emerald Group Publishing Limited
0957-4093
DOI 10.1108/IJLM-05-2013-0064

or goods from one source to a destination. The multimodal transport problem has been addressed by many authors proposing different abstractions and algorithms (Banomyong and Beresford, 2001; Qu and Chen, 2008; Zhang *et al.*, 2010; Ayed *et al.*, 2011; Kengpol *et al.*, 2012a, b). The previous research study found that the selected multimodal transportation routes have emphasized the selection of a multimodal transportation route for minimum cost and time: however, there appears to be only a few researches dealing with minimum risk and environmental setting (Min, 1991; Southworth and Peterson, 2000; Banomyong and Beresford, 2001; Ham *et al.*, 2005; Chang, 2008). In the case of transportation and logistics processes, accidents and risks not only imply a direct cost, but also reduce the competitiveness of exports (The UNCTAD Secretariat, 2003). At present, there are a lot of researches on transportation risk assessment because an accident may arise unexpectedly at any point along the way and transportation risks have a critical effect on the quality of transportation but there is still no research on the multimodal transportation yet. The previous research study found that the transportation risk assessments have emphasized the risk analysis of hazardous material or dangerous goods transportation (Scenna and Cruz, 2005; Verma, 2011; Reniers and Dullaert, 2013). Furthermore, several researchers have studied only risk on road, rail, air or waterway transport but have not studied risk on multimodal transportation in the research; for example, Tsai and Su, 2004; Scenna and Cruz, 2005; Verma, 2011. Additionally, Banomyong and Beresford (2001) have considered risk of route but do not combine it in the model. Therefore, the new conceptual framework for quantitative risk assessment (QRA) in multimodal transportation from the points of view of both logistics service providers (LSPs) and customers is proposed to combine into the model of this research.

Importantly, environmental issues in logistics and the supply chain are currently in place in various regions of the world. Proactive environmental management can be essential for the transportation industry and for organizations managing multimodal transportation hubs that integrate several types of freight carriers, logistics services and manufacturing or processing activities at a single site (Bielli *et al.*, 2006). A review to the previous studies reveals that there have been a lot of researchers who tried to reduce the volume of CO_2 emission; for example, Timilsina and Shrestha, 2009; Hao *et al.*, 2012. Therefore, environmental aspects are integrated with the model in this research.

Thus, the objective of this research is to develop a framework for route selection in multimodal transportation which can optimize cost, lead time, risk and CO_2 emission in multimodal transportation systems. This research proposes the development of a framework for route selection in multimodal transportation that includes a six-phase framework to select an optimal multimodal transportation route. The first phase is to collect the data of each route and select the origin and destination. The second phase is to calculate time and cost of each route by using multimodal transport cost-model. In the third phase, the CO_2 emissions are calculated based upon the 2006 guidelines of the Intergovernmental Panel on Climate Change (IPCC). The fourth phase proposes an integrated QRA, analytic hierarchy process (AHP) and data envelopment analysis (DEA) methodology to evaluate multimodal transportation risk. The fifth phase is to prioritize criteria by using the AHP which can be used in the objective function. The final phase is to optimize the route by using the zero-one goal programming. The aims of the model are to minimize transportation costs, transportation time, risk and CO_2 emission. The approach has been tested on a realistic multimodal

transportation service, originating from Bangkok in Thailand to a destination at Da Nang port in Vietnam.

This research is organized into four sections and the structure of this paper is as follows. After an introduction in Section 1, literature reviews and the development of a framework for route selection in multimodal transportation are presented in Section 2. Section 3 gives an illustrative example with a user in a transportation route between Bangkok, Thailand and Da Nang port, Vietnam. Finally, Section 4 presents conclusions, recommendations and an outlook on future research.

2. Literature reviews and the development of a framework for route selection in multimodal transportation

The objective of this research is to develop a framework for route selection in multimodal transportation which reduces cost, lead time, risk and CO_2 emission in multimodal transportation systems. The multi-criteria should be included to achieve an optimal multimodal transportation route. The aims of the model are to minimize transportation costs, transportation time, risk and CO_2 emission. This new conceptual framework for route selection in multimodal transportation consists of a six-phase framework to select an optimal multimodal transportation route which has been illustrated in Figure 1.

2.1 Phase I: scope definition

In the first phase, the user defines origin and destination and prepares the data of each route in route selection that consists of a possible multimodal transportation route for origin and destination. These data can be collected from interview and brainstorming of experts and LSPs.

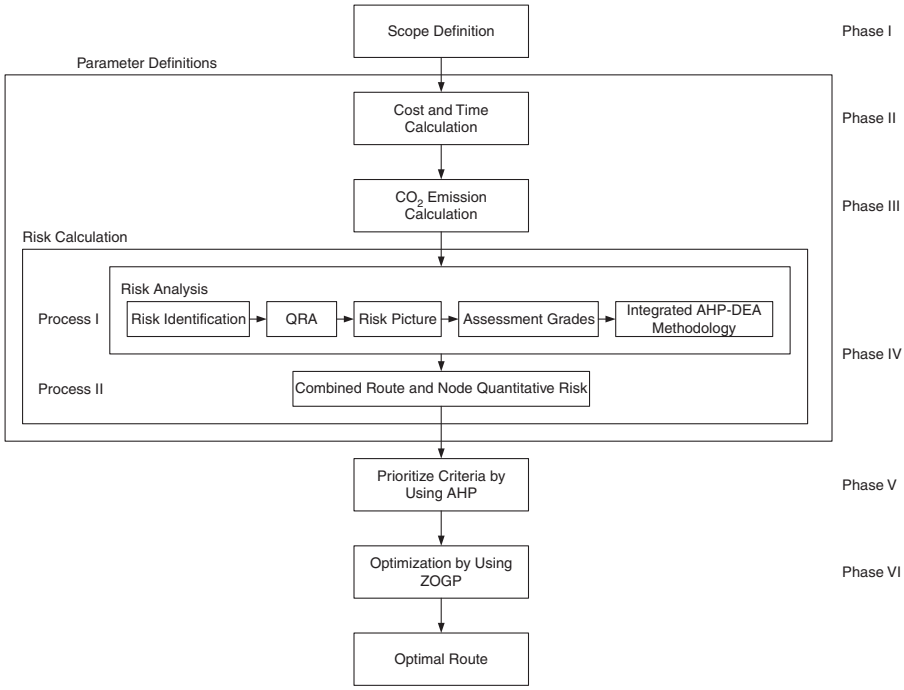


Figure 1.
The new conceptual framework for route selection in multimodal transportation

2.2 Phase II: cost and time calculation

The selection of a transport mode or combination of transport mode has a direct impact on transportation cost. Then, the objective of this phase is to propose a methodology about the cost and time components of a multimodal transportation. There are internal and external costs associated with a multimodal transportation. Internal costs consist of transport cost, time cost (intermodal transfer) and handling cost (Banomyong and Beresford, 2001; Janic, 2007). The external costs include the cost of the impacts of both environment and risk or accident. In this research, transportation cost and time for each route can be calculated by using Equation (1)-(4) which has been adapted from Janic (2007). After that, the multimodal transport cost model is used to calculate the transportation cost and time for each possible multimodal transportation route which has been adapted from Beresford and Dubey in 1990 and developed by Beresford in 1999, as improved by Banomyong and Beresford (2001):

$$\text{Transport Cost} = \left[\frac{(\text{Demand})}{(\text{Load Factor}) \times (\text{Vehicle Capacity})}\right] \times (\text{Cost per Frequency}) \quad (1)$$

$$\text{Time Cost} = (\text{Demand}) \times (\text{Time}) \times (\text{Cost per unit of time per unit of demand}) \quad (2)$$

$$\text{Handling Cost} = (\text{Demand}) \times (\text{Cost per unit of demand}) \quad (3)$$

$$\text{External Cost} = \left[\frac{(\text{Demand})}{(\text{Load Factor}) \times (\text{Vehicle Capacity})}\right] \times (\text{External Cost per frequency}) \quad (4)$$

2.3 Phase III: CO_2 emission calculation

The environmental problems and the climate changes are some of the controversial issues in society and life. Due to population increases, economic development and human activities, the atmospheric concentration of carbon dioxide have been rising extensively which leads to climate change and global warming. The increasing of average world temperature has a direct impact on the increase in greenhouse gases, and the increased contribution of carbon dioxide emission has an effect on the problem of global warming (Yabe, 2004). The previous research studies found that the transportation sector has been the second largest source of anthropogenic carbon dioxide emissions. Since the 1990s, the CO_2 emissions from the transport sector have grown rapidly and is expected to increase by 50 percent in 2030 (International Energy Agency, 2009). Road transport accounts for 74 percent of the transport sector's carbon dioxide emissions. Marine shipping produces 14 percent of all transport carbon dioxide emissions. Flight transport accounts for 11 percent of all transport carbon dioxide emissions. A review to the previous studies reveals that there have been a lot of researchers which tried to reduce the volume of CO_2 emission; for example, Timilsina and Shrestha, 2009; Hao *et al.*, 2012. Therefore, this research focusses on the environmental impact (EI) of carbon dioxide emission according to the Kyoto Protocol (Kengpol *et al.*, 2012b).

The CO_2 emissions can be calculated by summation of multiplying the amount of energy consumption by the CO_2 emission factor, with reference to the estimation

methodologies and CO_2 emission factors prescribed in the 2006 Guidelines of the IPCC and Jirajariyawet (2010), using the following calculation formula: (Eggleston *et al.*, 2006):

$$CO_2\,\text{Emission} = \sum \left(EF_{\text{Fuel}} \times FC_{\text{Fuel}}\right) \tag{5}$$

or

$$CO_2\,\text{Emission} = \sum \left[\left(EF_{\text{Transport}}\right) \times (\text{Distance} \times \text{Load})\right] \tag{6}$$

where EF_{Fuel} is the CO_2 emission coefficient of each fuel type (Emission Factor) which reference from Eggleston *et al.* (2006); FC_{Fuel} the amount of utilization of each fuel type (Fuel Consumption); and $EF_{\text{Transport}}$ the CO_2 emission coefficient of each transport type (Emission Factor).

The CO_2 emission coefficient of each transport type (Emission Factor) has been illustrated in Table AI. (More details of Table AI can be seen in Appendix 1.) The calculation is based upon scientific data for default vehicles and load factors (Kengpol *et al.*, 2012b). In this phase, the CO_2 emission can be calculated by using Equation (6). After that, the results of these calculations (CO_2 emission) are converted into an EI index form. The EI index is based upon a five-point scale in Table I. In this study, the EI index is derived from interviewing environmental experts.

2.4 Phase IV: risk calculation

This phase is the risk calculation process. There are two processes in this phase. The first process is the risk analysis. The second process is a combination between route and node quantitative risk. More details of this phase can be seen as follows:

Process I: risk analysis. In this study, this process proposes an integrated QRA, AHP and DEA methodology to evaluate multimodal transportation risk. According to Figure 1, the risk analysis process has five steps.

The first step is risk identification. The analysis of the nature of multimodal transport risk is based upon the failure mode and effect analysis (FMEA) methodology. FMEA was developed as a formal designed methodology in 1963 by the aerospace industry (Liu *et al.*, 2013), and has been studied extensively and used in a number of applications relating to assessing potential failures and preventing them (Liu *et al.*, 2013). FMEA can be applied for assessing the risk score of the occurrence, detection and severity of failure modes, each on a scale of 1-10 (see in Appendix 2.) and calculating the risk priority number (RPN) (see in Equation (7)) to select failure modes in multimodal transportation:

$$\text{RPN} = O \times S \times D \tag{7}$$

where O is the probability of the failure, S is the severity of the failure, and D is the probability of not detecting the failure. The failures modes with the highest RPN scores

Environmental impact index	CO_2 emission ($kgCO_2e$)	Description
1	0-100	Almost no impact
2	101-1,000	Not high impact
3	1,101-2,000	Fairly high impact
4	2,001-3,000	Impact
5	More than 3,000	Very high impact

Table I.
Environmental impact index scale

are the most urgent for consideration and improvements to reduce these risks in multimodal transportation. This study, the highest RPN scores analysis by using the 80/20 rule to Pareto scores is used in the next step. The failures modes or risk factors in multimodal transportation are obtained from previous researches and information from the LSPs interviews.

The second step is a quantitative risk analysis process. The QRA is used to determine the risk level of an activity by which people, environment or system might be in hazard. In transportation risk assessment, quantitative risk can be calculated by multiplying the probability of accident occurrence by the accident consequence as indicated in Equation (8) (Tsai and Su, 2004; Soons *et al.*, 2006):

$$R = P \times C \tag{8}$$

where R is risk level, P is the probability or frequency of accident occurrence, C is the consequences of the accident.

In the third step, a diagram is employed to map a transportation risk and to define a set of assessment grading in linguistic terms. A diagram can be called a risk picture, indicating its probability on the vertical axis and its potential severity on the horizontal axis, which have been shown in Figure 2.

The fourth step is the assessment grades in linguistic terms from a risk picture in the previous step. This technique is also known as semi-quantitative risk analysis (Vose, 2008). The set of assessment grades has been defined from a risk picture as follows, squares with scores >13 define the high-risk region (H), squares with scores between 7 and 12 define the medium-risk region (M), squares with scores between 3 and 6 define the low-risk region (L) while squares with scores smaller than 3 define the none-risk region (N).

The final step is the process of a new integrated AHP-DEA methodology. DEA is the method for evaluating the efficiency of homogeneous units with multiple inputs and multiple outputs which is developed by Charnes *et al.* in 1978 (Charnes *et al.*, 1978; Soltanifar and Lotfi, 2011). This method is a linear programming method that can be used for measuring the relative efficiencies of decision-making units (DMUs) (Charnes *et al.*, 1978; Ramanathan, 2001; Mohajeri and Amin, 2010). In recent years, DEA has

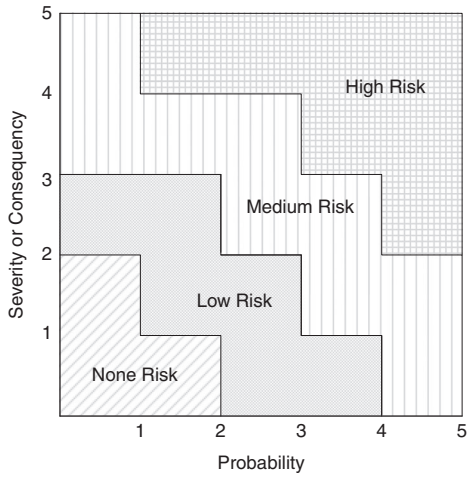


Figure 2.
Risk picture

been addressed by several successful applications for evaluation of many different kinds of DMUs. Furthermore, DEA technique is used for obtaining the weights of the pairwise comparisons matrix in the AHP method (Mohajeri and Amin, 2010). The previous research study found that the DEA is one of the most popular tools for performance measurement, while the AHP is a popular tool for solving a multi-criteria decision making problem (Ramanathan, 2006). There have been several ideas of combining the AHP and DEA because the AHP can only compare a very limited number of decision alternatives and it is also not easy when there are many alternatives to be ranked (Ramanathan, 2001; Liu and Hai, 2005; Wang *et al.*, 2008). Therefore, a new integrated AHP and DEA methodology is proposed to this problem for arranging a large number of decision alternatives. A new integrated AHP and DEA methodology in this paper is adapted from the paper of Ramanathan (2006), Wang *et al.* (2008), Soltanifar and Lotfi (2011). However, this research proposes an integrated QRA with a new integrated AHP and DEA methodology. In this step, there are four steps in the procedure of this new methodology. The new methodology has been illustrated in Figure 3.

Firstly, the AHP method is used to determine the weights of criteria. Next, experts are interviewed to collect data for calculating the risk level and assessing decision alternatives by using the defined assessment grades in linguistic terms with respect to each criterion (Wang *et al.*, 2008). Then, the DEA method is used to determine the values of the defined assessment grades in linguistic terms and to generate the local risk scores. The DEA model with common weights which is adapted from the paper of Ramanathan (2006), Wang *et al.* (2008), Soltanifar and Lotfi (2011), is as follows:

$$\begin{aligned} &\text{Maximize} \quad \alpha \\ &\text{Subject to} \quad \alpha \leqslant v_{ij} = \sum_{k=1}^{K_j} S(L_{jk}) NE_{ijk} \leqslant 1, i = 1, \ldots, n \\ &\qquad S(L_{j1}) \geqslant 2S(L_{j2}) \geqslant \cdots \geqslant K_j S(L_{jK_j}) \geqslant 0 \end{aligned} \tag{9}$$

where $L_{j1} \ldots, L_{jK_j}$ represent the linguistic term of importance from the most to the least important, K_j is the number of assessment grades for criterion j, $NE_{ijk}(k = 1, \ldots, K_j)$ are the numbers of the experts who assess alternative A_i to Grade L_{jk} under the criterion j, $S(L_{j1}), \ldots, S(L_{jKj})$ are decision variables and $S(L_{j1}) \geqslant 2S(L_{j2}) \geqslant, \ \ldots, \ \geqslant K_j S(L_{jkj}) \geqslant 0$ is the strong ordering condition imposed on assessment grades (Noguchi *et al.*, 2002; Liu and Hai, 2005; Wang *et al.*, 2008). Therefore, the local weight of each alternative with respect to every criterion can be defined as follows (Wang *et al.*, 2008):

$$v_{ij} = \sum_{k=1}^{K_j} S(L_{jk}) NE_{ijk}, for\ i = 1, \ldots n; \quad j = 1, \ldots, m \tag{10}$$

Finally, the local risk scores are aggregated into an overall risk score for each decision alternative by using the simple additive weighting (SAW) method. The SAW is a



Figure 3.
The procedure of a new integrated AHP-DEA methodology

weighted linear combination or scoring method which is a simple method and most often used as a multi attribute decision technique (Afshari *et al.*, 2010). The decision maker can obtain an overall weight for each alternative simply by multiplying the scale rating for each attribute value by the importance weight assigned to the attribute and then summing these products over all attributes (Tillman and Cassone, 2012). The SAW model can be formulated as follows in Equation (11):

$$V(A_i) = \sum_{j=1}^{m} w_j v_{ij}^* = \sum_{j=1}^{m} w_j \left(\sum_{k=1}^{K_j} S^*(L_{jk}) NE_{ijk} \right), i = 1, \ldots, n \tag{11}$$

where w_j are the criteria weights determined by the AHP methodology, $S^*(L_{jk})$ are the optimal scorings of the assessment grades determined by Equation (9) and $V(A_i)$ are the overall weights of the n decision alternatives, that the n alternatives can be ranked (Wang *et al.*, 2008).

Process II: combination between routes and nodes quantitative risk. This process is the second process of risk calculation. After, the decision maker calculates an overall risk score for each decision alternative. The decision maker has to combine routes and nodes quantitative risk score for each possible multimodal transportation route. In general, the multimodal transportation consists of routes, where the goods are in motion by different modes of transportation and nodes, where goods come to rest or change another mode of transport. The causes of risk at transportation nodes can differ from the causes that occur along the route of the transport chain. Therefore, this process is to combine routes and nodes quantitative risk score of multimodal transportation by using a combined routes and nodes quantitative risk model, which has been shown as follows:

$$\text{Risk along route} = \sum_{i=1}^{n} A_i \tag{12}$$

where A_i are the possible decision alternative of multimodal transportation route.

2.5 Phase V: prioritize criteria by using AHP methodology

In this phase, the significant weight of each criterion for each transportation situation is obtained by using AHP methodology. The AHP has been widely applied in several previous studies to solve a multi-criteria decision problem. The AHP was developed by Saaty in 1980 (Saaty, 1990). The AHP method decomposes a complex MCDM problem into a graphical hierarchy with respect to one or more criteria. The AHP method provides a structured framework for setting priorities on each level of the hierarchy using pairwise comparisons that are quantified using 1-9 scales (Saaty, 1990). These weights are integrated into the objective function of zero-one goal programming. The procedure of AHP can be summarized as: first, a complex MCDM problem is decomposed into a graphical hierarchy with respect to one or more criteria. The hierarchical structure is based upon the AHP methodology that contains the decision goal, the alternatives for reaching the goal and the criteria for assessing the alternatives, which have been illustrated in Figure 4.

Second, the significant weight of each criterion is assigned by pairwised comparison. Third, the pairwise comparison matrix A should have an acceptable consistency, which

can be checked by following the consistency ratio (CR). Finally, the new conceptual framework for route selection in multimodal transportation uses the significant weight of each criterion for optimization by using ZOGP methodology in the next phase.

2.6 Phase VI: optimization by using ZOGP methodology

The last phase is the ZOGP methodology to optimize multimodal transportation route. In order to achieve this, the problem is generally formulated by using the ZOGP model. The significant weight from AHP, parameters and limited data from previous phases are used to formulate the objective function and constraints. The model of integrated AHP and ZOGP is presented as following:

$$\text{Minimize} = \sum_{i=1}^{m} \left(g_i d_i^+\right) = w_1\left(g_1 d_1^+\right) + w_2\left(g_2 d_2^+\right) + \cdots w_m\left(g_m d_m^+\right) \tag{13}$$

$$\text{Subject to} \quad \text{Budget} \qquad c_1X_1 + c_2X_2 + \cdots + c_nX_n \leqslant C \tag{14}$$

$$\text{Time} \qquad t_1X_1 + t_2X_2 + \cdots + t_nX_n \leqslant T \tag{15}$$

$$\text{Environmental Impact} \qquad e_1X_1 + e_2X_2 + \cdots + e_nX_n \leqslant E \tag{16}$$

$$\text{Risk} \qquad r_1X_1 + r_2X_2 + \cdots + r_nX_n \leqslant R \tag{17}$$

$$\begin{aligned} &X_1 + X_2 + \cdots + X_n = 1 \\ &w_i d_i^+ \geqslant 0, \textit{ for } i = 1, 2, \ldots, m \\ &c_j, t_j, e_j r_j \geqslant 0, \textit{ for } j = 1, 2, \ldots, n \\ &X_j = 0 \textit{ or } 1, \textit{ for } j = 1, 2, \ldots, n \end{aligned} \tag{18}$$

In this research, there are four objective functions (or goals). The first objective is the transport budget (g_1) that is not over the user limit. The second objective is the transport time (g_2) that is not over the lead time limited by user. The third objective is EI (g_3) that is less than the minimum scale which the user limits. Finally, the fourth objective is the multimodal transportation risk (g_4) that is less than the user limit. All data in each objective function of this research have a different unit therefore a converting unit is needed by formulating all units to percentage.

The Equations (13)-(18) can be defined by the deviation variables, decision variables and parameters as below. Equation (18) controls that only one route is optimum for one situation (Kengpol *et al.*, 2012b).

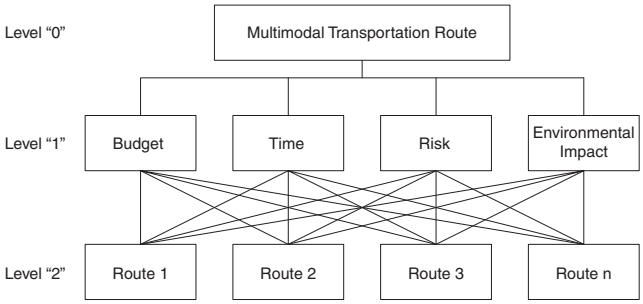


Figure 4.
The AHP structure for choice multimodal transportation route

Deviation variables

d_i^+ = The overachievement of goal i.

Decision variables. X_j = Zero-one variables representing the non-selection (i.e. a zero) or selection (i.e. a one) of route $j = 1, 2, 3, \ldots, n$, subject to criteria constraint in the right hand side (budget, time, EI and risk).

Parameters. w_i = The weights for each criterion coefficients relating to deviation of each goal that can obtain from AHP. (w_1 is weight of budget criterion, w_2 is weight of time criterion, w_3 is weight of EI criterion and w_4 is weight of risk criterion).

c_j = The coefficient of X_j in budget constraint that is a cost of each route in percentage of the user budget:

$$c_j = \left[\frac{(\text{Budget limited by user}) - (\text{Cost of route}j)}{(\text{Budget limited by user})}\right] \times 100$$

C = The right hand side of Equation (14) is percentage of budget limited by user that is presented below:

$$C = \left[\frac{(\text{Budget limited by user}) - (\text{Minimum cost of all route})}{(\text{Budget limited by user})}\right] \times 100$$

t_j = The coefficient of X_j in transport time constraint that is a percentage of transport time of each route which is limited by user:

$$t_j = \left[\frac{(\text{Transport time of route}j)}{(\text{Transport time limited by user})}\right] \times 100$$

T = The right hand side of Equation (15) is percentage of transport time limited by user (= 100%).

e_j = The coefficient of X_j in EI constraint:

$$e_j = \left[\frac{(\text{EI scale limited by user}) - (\text{EI risk scale of route}j)}{(\text{EI scale limited by user})}\right] \times 100$$

E = The right hand side of EI constraint in Equation (16):

$$E = \left[\frac{(\text{EI scale limited by user}) - (\text{Minimum EI scale of all route})}{(\text{EI scale limited by user})}\right] \times 100$$

r_j = The coefficient of X_j in multimodal transportation risk constraints:

$$r_j = \left[\frac{(\text{MTrisk limited by user}) - (\text{MT risk of route}j)}{(\text{MT risk limited by user})}\right] \times 100$$

R = The right hand side of multimodal transportation risk constraint in Equation (17):

$$R = \left[\frac{(\text{MT risk limited by user}) - (\text{Minimum MT risk of all route})}{(\text{MT risk limited by user})}\right] \times 100$$

where *EI* scale limited by the user is defined as the maximum acceptable EI scale which is limited by the user or the decision maker. More details of the EI index scale can be seen in Table I, as described earlier.

3. A case study between Thailand and Vietnam

This section demonstrates the application of the proposal for a new conceptual framework for route selection in multimodal transportation: a case study is conducted on a realistic multimodal transportation service.

Since The Greater Mekong Sub-Region was mentioned as the region with the most rapid economic development in Asia, Thailand and Vietnam have been increasing trade volumes of various kinds of goods every year as a result of the opened economy. This case is to study a transport service, originating from Bangkok in Thailand to the destination in Da Nang in Vietnam. The database in this research is derived from collecting data through interviews with both LSPs and customers. This case study is based upon the conceptual framework for route selection in multimodal transportation which has been illustrated in Figure 1, as described earlier.

3.1 Scope definition of case study

This phase is the definition of the original and destination of transport service and possible multimodal transportation route. There are 11 multimodal transportation routes which originate from Bangkok in Thailand to the destination in Da Nang in Vietnam. The 11 multimodal transportation routes have been illustrated in Table II.

Route	Segmented route 1	2	3	4	5	6
1. BKK###HMC—Da Nang	A_1	A_2	A_3	A_4	A_5	A_6
2. BKK###HMC +++ Da Nang	A_1	A_2	A_3	A_7	A_8	A_9
3. BKK***Da Nang	A_{10}	A_{11}	A_{12}			
4. BKK###Da Nang	A_1	A_{13}	A_{14}			
5. BKK—HMC###Da Nang	A_{15}	A_{16}	A_4	A_3	A_{17}	A_{14}
6. BKK—HMC +++ Da Nang	A_{15}	A_{16}	A_4	A_7	A_8	A_9
7. BKK—HMC***Da Nang	A_{15}	A_{16}	A_4	A_{18}	A_{19}	A_{12}
8. BKK***HMC +++ Da Nang	A_{10}	A_{20}	A_{18}	A_7	A_8	A_9
9. BKK***HMC—Da Nang	A_{10}	A_{20}	A_{18}	A_4	A_5	A_6
10. BKK***HMC###Da Nang	A_{10}	A_{20}	A_{18}	A_3	A_{17}	A_{14}
11. BKK###HMC***Da Nang	A_1	A_2	A_3	A_{18}	A_{19}	A_{12}

Notes: A_1, Bangkok (BKK) port; A_2, The route from Bangkok to Ho Chi Minh City (HCM) by sea transport mode (BKK ### HMC); A_3, Ho Chi Minh City (HMC) port; A_4, Ho Chi Minh City (HMC) station; A_5, The route from Ho Chi Minh City (HMC) to Da Nang by truck transport mode (HMC — Da Nang); A_6, Da Nang station; A_7, Ho Chi Minh City (HMC) railroad station; A_8, The route from Ho Chi Minh City (HMC) to Da Nang by train transport mode (HMC +++ Da Nang); A_9, Da Nang railroad station; A_{10}, Bangkok (BKK) airport; A_{11}, The route from Bangkok to Da Nang by air transport mode (BKK ***Da Nang); A_{12}, Da Nang airport; A_{13}, The route from Bangkok to Da Nang by sea transport mode (BKK ### Da Nang); A_{14}, Da Nang port; A_{15}, Bangkok (BKK) station; A_{16}, The route from Bangkok to Ho Chi Minh City (HMC) by truck transport mode (BKK — HMC); A_{17}, The route from Ho Chi Minh City (HMC) to Da Nang by sea transport mode (HMC###Da Nang); A_{18}, Ho Chi Minh City (HMC) airport; A_{19}, The route from Ho Chi Minh City (HMC) to Da Nang by air transport mode (HMC***Da Nang); A_{20}, The route from Bangkok to Ho Chi Minh City (HMC) by air transport mode (BKK***HMC). —, truck; +++, train; ***air and ###, sea

Table II.
The possible multimodal transportation route

In this research, the selection of an optimal multimodal transportation route is based upon the multi-criteria that consist of transportation cost, transportation time, EI and transportation risk.

3.2 Transportation cost and time calculation

From the possible multimodal transportation route in the previous phase, the selection of transport mode or combination of transport modes has a direct impact on transportation cost and transportation time. Transportation cost and time for each possible multimodal transportation route can be calculated by using Equation (1)-(4) which have been illustrated in Table III. The database in this research is derived from collecting data through interviews with both LSPs and customers.

3.3 CO_2 emission calculation

In the third phase, the EI is considered in CO_2 emission term. The 11 multimodal transportation routes have been divided into nine segmented routes. The database for CO_2 emission calculation has been shown in Table IV.

The volume of CO_2 emission can be calculated by using Equation (6) which is based on scientific data in Tables AI and IV for default vehicles and load factors. The volume of CO_2 emission calculation for each segmented route has been shown in Table V.

Route	Transportation cost (USD/TEU)	Transportation time (days)
1. BKK###HMC—Da Nang	2,967.5	8
2. BKK###HMC +++ Da Nang	1,631.5	10
3. BKK***Da Nang	4,580	2
4. BKK###Da Nang	1,080	8
5. BKK—HMC###Da Nang	4,298	7
6. BKK—HMC +++ Da Nang	4,254	7
7. BKK—HMC***Da Nang	7,880	4
8. BKK***HMC +++ Da Nang	5,334	6
9. BKK***HMC—Da Nang	6,580	4
10. BKK***HMC###Da Nang	5,378	6
11. BKK###HMC***Da Nang	5,257.5	7

Notes: TEU, Twenty-Foot Equivalent Units; —, truck; +++, train; ***air and ###, sea

Table III.
Database of transportation cost and transportation time

Segmented route		Load (ton)	Distance (km)
A_2	BKK ### HMC	16	1,180
A_5	HMC — Da Nang	16	947
A_8	HMC +++ Da Nang	16	935
A_{11}	BKK ***Da Nang	16	1,200
A_{13}	BKK ### Da Nang	16	1,600
A_{16}	BKK — HMC	16	1,100
A_{17}	HMC###Da Nang	16	1,000
A_{19}	HMC***Da Nang	16	1,000
A_{20}	BKK***HMC	16	1,000

Notes: —, Truck; +++, train; ***air and ###, sea

Table IV.
Database of transportation cost and transportation time

Segmented route		CO_2 emission ($kg \cdot CO_2e$)
A_2	BKK ### HMC	188.8
A_5	HMC — Da Nang	716.6896
A_8	HMC +++ Da Nang	1,662.056
A_{11}	BKK ***Da Nang	3.84
A_{13}	BKK ### Da Nang	256
A_{16}	BKK — HMC	832.48
A_{17}	HMC###Da Nang	160
A_{19}	HMC***Da Nang	3.2
A_{20}	BKK***HMC	3.2

Notes: —, Truck; +++, train; ***air and ###, sea

Table V.
The volume of CO_2 emission calculation for each segmented route

For example, according to Table V, the segmented route A_2 (BKK ### HMC) has the volume of CO_2 emission 188.8 kg.CO_2e which can be calculated by using Equation (6). The one originating from Bangkok in Thailand to the destination in Ho Chi Minh City (HMC) by sea transport mode has a distance of 1,180 km and capacity loads 16 tons. The CO_2 emission coefficient of container ship transport mode is 0.01 $kg \cdot CO_2e$/ton-km. Then, the volume of CO_2 emission of the segmented route A_2 (BKK ### HMC) is $(0.01) \times (1{,}180 \times 16) = 188.8\,kg \cdot CO_2e$.

Then, the volume of CO_2 emission of the 11 possible multimodal transportation routes are calculated by combination between two possible segmented routes in Table V which can be followed by the 11 possible multimodal transportation routes in Table II. The volumes of CO_2 emission of the 11 possible multimodal transportation routes are presented in column 2 of Table VI. After that, the volume of CO_2 emission is converted into an EI index form. The EI index is based upon a five point scale in Table I. The EI index for each possible multimodal transportation route is shown in column 3 of Table VI.

3.4 Transportation risk calculation

The fourth phase is the transportation risk calculation process. The explanation of procedure can be seen in more detail in Section 2.4, as described earlier.

Route	CO_2 emission ($kg \cdot CO_2e$)	Environmental impact scale
1. BKK###HMC—Da Nang	905.49	2
2. BKK###HMC +++ Da Nang	1,850.86	3
3. BKK***Da Nang	3.84	1
4. BKK###Da Nang	256	2
5. BKK—HMC###Da Nang	992.48	2
6. BKK—HMC +++ Da Nang	2,494.54	4
7. BKK—HMC***Da Nang	835.68	2
8. BKK***HMC +++ Da Nang	1,665.26	3
9. BKK***HMC—Da Nang	719.89	2
10. BKK***HMC###Da Nang	163.2	2
11. BKK###HMC***Da Nang	192.00	2

Notes: —, Truck; +++, train; ***air and ###, sea

Table VI.
Database of environmental impact scale

Process I: risk analysis. In this study, the risk analysis has five steps. The first step is risk identification. The analysis of the nature of multimodal transport risk is based on FMEA methodology. This study adopts the risk factors in previous researchers (Kengpol *et al.*, 2012a; Vilko and Hallikas, 2012) and information from the LSPs interviews. The risk factors can be assessed in terms of the following criteria:

(1) *Freight damaged risks* are defined as the situation of loss of goods during transfer mode, damaged freight, damaged from transportation, damaged from delivery at warehouse, damaged from delivery to customer.

(2) *Infrastructure and equipment risks* are defined as capacity of bridges on the route, tunnels, slope and the width of roads, capacity of port, risk of shipment in the rainy season or in a tropical storm, accident rates and facility of equipment material handling and capacity problem in railroad traffic.

(3) *Political and legislatorial risks* are defined as laws, political risk, traffic rules and customs.

(4) *Operational risks* are defined as lack of skilled workers, standardization of documents, interpretation problems with documents or contracts or permits, ship collision or sinking in the approach lane.

(5) *Macro risks* are defined as financial crisis, fierce competition in the transportation sector and unattractive markets.

(6) *Environment risks* are defined as climate change, toxic waste and the (aesthetic) appearance of the route or building.

The risk factors or failures modes are subjected to traditional FMEA by using categorical rankings for occurrence (O), severity (S) and detection (D), to a scale from 1 to 10 (Tables AII-AIV). For each of the risk factors or failures modes, RPN values are calculated by Equation (7). Six potential failure modes are identified by the FMEA team that consists of ten decision makers from several expert groups. The ten decision makers assess their judgments on the six failure modes with occurrence (O), severity (S) and detection (D). The results of this traditional FMEA are shown in Table VII.

From Table VII, the results show that the freight damaged risks have the highest risk (RPN = 392), which should be prioritized for inspection. The overall risk priority ranking of six categories of risk factors is assessed as freight damaged risks > infrastructure and equipment risks > operational risks > political and legislatorial risks > macro risks > environment risks. This study selects the first four highest RPN scores to use in the next step that consists of freight damaged risks, infrastructure and equipment risks, operational risks, and political and legislatorial risks. The first four

Table VII.
The results of this traditional FMEA for analysis of the multimodal transport risk

Risk factor	Traditional FMEA (O)	(S)	(D)	RPN
Freight damaged risks	7	7	8	392
Infrastructure and equipment risks	7	8	6	336
Political and legislatorial risks	8	9	3	216
Operational risks	6	6	7	252
Macro risks	5	5	8	200
Environment risks	4	4	8	128

highest RPN scores are obtained by using the 80/20 rule of Pareto which can be illustrated in Figure 5.

The second step is the quantitative risk analysis process which can be assessed for risk level by multiplying the probability factor by the severity as shown in Equation (8). In this study, the probability factor and the severity are formulated as rank scale. The identification rank of probability assessment scale and rank of severity impact assessment scale are adopted from Hallikas *et al.* (2004) and Kengpol *et al.* (2012a).

The rank of probability assessment scale of freight damaged risks is identified by using percentage freight damaged and loss of information and the severity impact assessment scale are identified by the percentage of damaged goods value and total goods value (Kengpol *et al.*, 2012a) which have been illustrated in Table VIII. In infrastructure and equipment risk, the rank of probability assessment scale is identified by using percentage of accident rate and the rank of severity impact assessment scale of risk is identified by using a percentage of increased time of transportation on the route (Kengpol *et al.*, 2012a) which has been illustrated in Table IX. In operational risk, the rank of probability assessment scale is identified by using percentage of occurrence rate, and the rank of severity impact assessment scale of risk is identified by using percentage of increased time of transportation on the route (Kengpol *et al.*, 2012a) which has been illustrated in Table IX. The last risk factor, the

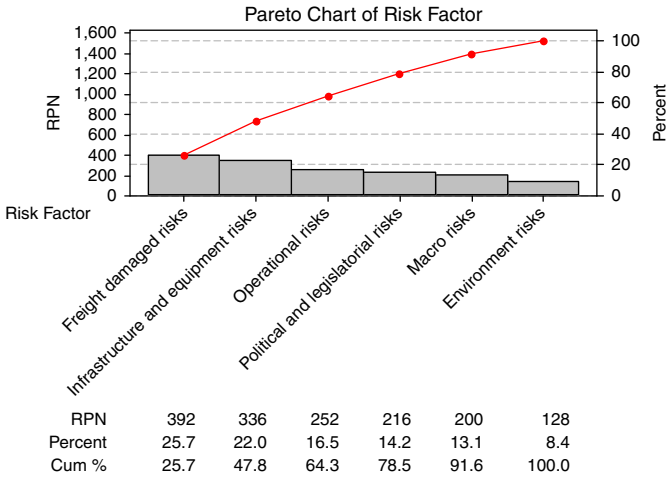


Figure 5.
The 80/20 rule to Pareto for first four highest RPN scores analysis

Rank	Probability assessment scale of freight damaged risks (%)	Severity impact assessment scale of freight damaged risks (%)
1	0.00-0.01	0-5
2	0.02-0.10	6-10
3	0.11-1.00	11-15
4	1.01-5.00	16-20
5	More than 5.00	More than 20

Sources: Hallikas *et al.* (2004) and Kengpol *et al.* (2012a)

Table VIII.
The rank of probability and severity impact assessment scale of freight damaged risks

rank probability assessment scale of political and legislatorial risks are identified by using percentage of occurrence rate, and the rank of severity impact assessment scale of risk is identified by using percentage of increased time of transportation on the route (Kengpol *et al.*, 2012a) which has been illustrated in Table IX as well.

In the third step, the risk picture is employed to map a transportation risk level and to define a set of assessment grades in linguistic terms. From the previous step, the product of the scores expressing in the likelihood of failure and the consequence from failure gives the risk score. The risk score can be measured by the risk magnitude in Figure 2 and can be used to define a set of assessment grades in linguistic terms in the next step. The risk picture illustrates an overall view upon all risks and helps LSPs to focus on the most important risks.

In the fourth step, the risk picture is transformed into assessment grades in linguistic terms. This process is necessary to define a set of assessment grades in linguistic terms to quantitatively describe transportation risk. The set of assessment grades has been defined from a risk picture as follows (and also can be seen in Figure 2):

$$\text{AssessmentGrade}(G) = \{\text{High}, \text{Medium}, \text{Low}, \text{None}\}$$

where the squares with scores >13 define the high-risk region (H); the squares with scores between 7 and 12 define the medium-risk region (M); the squares with scores between 3 and 6 define the low-risk region (L); and the squares with scores smaller than 3 define the none-risk region (N).

The final step is the integrated AHP-DEA methodology. It has four criteria and 20 alternatives. The hierarchical structure for multimodal transportation risk assessment has been illustrated in Figure 6.

Rank	Probability assessment scale of freight damaged risks (% of accident or occurrence rate)	Severity impact assessment scale of freight damaged risks (% of increased time of transportation)
1	0.00-0.01	0-10
2	0.02-0.10	11-20
3	0.11-1.00	21-30
4	1.01-5.00	31-40
5	More than 5.00	More than 40

Sources: Hallikas *et al.* (2004) and Kengpol *et al.* (2012a)

Table IX.
The rank of probability and severity impact assessment scale of infrastructure and equipment risks

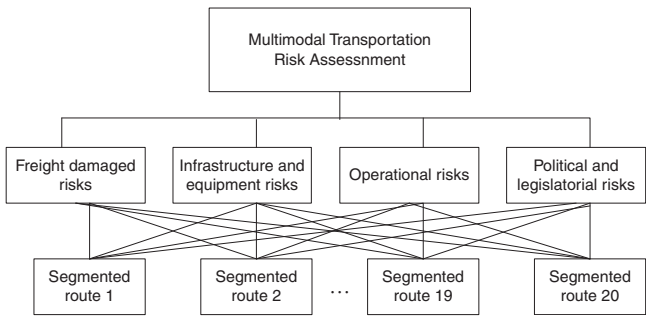


Figure 6.
Hierarchical structure for multimodal transportation risk assessment

In this step, there are four steps in the procedure of this new methodology. To start with, the LSPs have to determine the weights of criteria. The AHP method is used to determine the weights of criteria. Four risk assessment criteria are assessed by the AHP team. The AHP team comes from the FMEA team that consists of ten decision makers from several expert groups. The pairwise comparison matrix for the four risk assessment criteria provided by the AHP team is as follows (Table X).

The relative weight criteria from AHP are freight damaged risks 0.56, infrastructure and equipment risks 0.26, operational risks 0.12, and political and legislatorial risks 0.06 and the maximum eigenvalue (λ_{max}) is 4.18. The consistency index for the above paired comparison matrix (*CI*) is 0.059 and the corresponding *CR* is 0.066. Because the corresponding consistency is $<0.1 (CR<0.1)$, the pairwise comparison matrix is considered to have an acceptable consistency.

After defining and describing the set of assessment grades in the previous steps, the assessment of decision alternatives are begun in this step. Table XI shows the decision matrix of assessment results for the 20 segmented routes, which are assessed by ten experts or LSPs for each criterion.

The risk assessment data in Table XI is used to generate the local risk scores for each criterion by using Equation (9) based upon the DEA method. The local weight of each decision alternative with respect to every criterion is determined as a DMU,$S(L_{jk})$ as a decision variable and also the weight assigned to the output NE_{ijk}. The following optimal solutions of each criterion have been illustrated in Table XII.

These optimal solutions in Table XII can be used to calculate the local risk scores of the 20 segmented routes with respect to each of the four criteria by using Equation (10), which has been illustrated in the columns 2-5 of Table XIII. Finally, the local risk scores can be aggregated into an overall transportation risk score for each decision alternative by using the SAW method (see Equation 11). The overall transportation risk scores are presented in column 6 of Table XIII and the risk priorities ranking are shown in column 7 of Table XIII.

There is, for example, according to the segmented route A_{16}, the local risk score of each criterion which can be calculated as follows:

Freight damaged risks $[(9\times0.105263)+(1\times0.052632)]=0.999999$;

Infrastructure and equipment risks $[(7\times0.111111)+(2\times0.055556)+(1\times0.037037)]=0.925926$;

Operation risk $[(7\times0.105263)+(1\times0.052632)+(2\times0.35088)]=0.859649$

Political and legislatorial risk $[(5\times0.117647)+(5\times0.058824)]=0.882355$

The overall transportation risk score of the segmented route A_{16} is $[(0.56\times0.999999)+(0.26\times0.925926)+(0.12\times0.859649)+(0.06\times0.882355)]=0.9568394$. Similarly, the overall risk score of other segmented routes can be computed and ranked in Table XIII.

Process II: combination between routes and nodes quantitative risk. In order to obtain risk that occurs along the routes, a combined node quantitative risk model in Equation (12) has been conducted to combine routes and nodes quantitative risk score of multimodal transportation. The results of combination between routes and nodes quantitative risk are presented in Table XIV.

The results from Table XIV found that the highest overall transportation risk of the 11 multimodal transportation routes is Route 6 (BKK—HMC +++ Da Nang) and the smallest overall transportation risk of the 11 multimodal transportation routes is Route 3 (BKK***Da Nang). The overall transportation risk priority ranking of the 11 multimodal transportation routes are presented in column 4 of Table XIV. The optimal

Table X.
The pairwise comparison matrix for the four risk assessment criteria

	Freight damaged risks	*Infrastructure and equipment risks*	*Operational risks*	*Political and legislatorial risks*
A =				
Freight damaged risks	1	3	5	7
Infrastructure and equipment risks	1/3	1	3	5
Operational risks	1/5	1/3	1	3
Political and legislatorial risks	1/7	1/5	1/3	1
Consistency Matrix A =				
Freight Damaged Risks	1	3	5	7
Infrastructure and Equipment Risks	1/3	1	3	5
Operational Risks	1/5	1/3	1	3
Political and Legislatorial Risks	1/7	1/5	1/3	1
Summation in Column	1.68	4.53	9.33	16
Mean Matrix A =				
Freight Damaged Risks	1/1.68 = 0.60	3/4.53 = 0.66	5/9.33 = 0.54	7/16 = 0.44
Infrastructure and Equipment Risks	$\frac{1}{3}$/1.68 = 0.20	1/4.53 = 0.22	3/9.33 = 0.32	5/16 = 0.31
Operational Risks	$\frac{1}{5}$/1.68 = 0.12	$\frac{1}{3}$/4.53 = 0.07	1/9.33 = 0.11	3/16 = 0.19
Political and Legislatorial Risks	$\frac{1}{7}$/1.68 = 0.09	$\frac{1}{5}$/4.53 = 0.04	$\frac{1}{3}$/9.33 = 0.04	1/16 = 0.06
Weight Criteria =				
Freight Damaged Risks	= (0.60 + 0.66 + 0.54 + 0.44)/4	= 0.56		
Infrastructure and Equipment Risks	= (0.20 + 0.22 + 0.32 + 0.31)/4	= 0.26		
Operational Risks	= (0.12 + 0.07 + 0.11 + 0.19)/4	= 0.12		
Political and Legislatorial Risks	= (0.09 + 0.04 + 0.04 + 0.06)/4	= 0.06		

Segmented route	Assessment criteria: Freight damaged risks (10 experts)				Infrastructure and equipment risks (10 experts)				Operational risk (10 experts)				Political and legislatorial risk (10 experts)			
	H	*M*	*L*	*N*	*H*	*M*	*L*	*N*	*H*	*M*	*L*	*N*	*H*	*M*	*L*	*N*
A_1	5	5	0	0	2	3	4	1	0	3	7	0	0	0	4	6
A_2	5	4	1	0	4	3	3	0	1	2	7	0	0	0	9	1
A_3	2	3	4	1	1	3	5	1	9	1	0	0	7	2	1	0
A_4	9	1	0	0	4	3	2	1	3	5	1	1	3	4	2	1
A_5	8	2	0	0	8	2	0	0	4	4	2	0	7	3	0	0
A_6	4	3	3	0	1	1	7	1	2	5	3	0	2	2	5	1
A_7	7	2	1	0	2	6	1	1	7	1	1	1	1	8	1	0
A_8	6	4	0	0	3	3	4	0	6	2	2	0	1	4	4	1
A_9	5	4	1	0	0	8	2	0	0	6	4	0	0	0	6	4
A_{10}	1	2	2	5	1	1	7	1	1	1	8	0	3	0	7	0
A_{11}	1	1	1	7	0	0	1	9	2	3	2	3	1	1	7	1
A_{12}	2	2	2	4	0	0	9	1	4	2	2	2	1	4	1	4
A_{13}	6	3	1	0	2	2	6	0	2	1	7	0	2	6	1	1
A_{14}	4	4	2	0	6	2	2	0	1	5	2	2	0	3	7	0
A_{15}	3	3	4	0	5	2	2	1	5	2	3	0	2	2	4	2
A_{16}	9	1	0	0	7	2	1	0	7	1	2	0	5	5	0	0
A_{17}	7	2	1	0	2	2	6	0	5	4	1	0	0	4	5	1
A_{18}	1	2	1	6	3	2	4	1	8	2	0	0	1	2	3	4
A_{19}	1	1	4	4	2	3	3	2	3	2	2	3	0	0	3	7
A_{20}	0	2	2	6	1	1	3	5	2	1	4	3	0	0	2	8

Notes: *H*, High; *M*, medium; *L*, low; *N*, none

Table XI.
Risk assessment data for 20 segmented routes

Criteria	$s^*(H)$	$s^*(M)$	$s^*(L)$	$s^*(N)$	α^*
Freight damaged risks	0.105263	0.052632	0.035088	0.026316	0.333333
Infrastructure and equipment risks	0.111111	0.055556	0.037037	0.027778	0.287037
Operational risk	0.105263	0.052632	0.035088	0.011005	0.403509
Political and legislatorial risk	0.117647	0.058824	0.039216	0.029412	0.313725

Table XII.
The following optimal solutions of each criterion

route is air transport mode departing from Bangkok airport to Da Nang airport (Route 3) because the overall transportation risk score is always the-smaller-the-better (Wang *et al.*, 2008). Route 3 has a small overall transportation risk score but has a high cost. Then, the LSPs should consider together with other criteria to obtain the next appropriate route.

The data from Tables III, VI and XIV can be summarized into the overall database of multimodal transportation route which has been illustrated in Table XV.

3.5 Prioritized criteria by using AHP methodology

In this phase, the LSPs are asked to determine the significant weights of criteria for each transportation situation by using the AHP method. The objective of this research is to develop a framework for route selection in multimodal transportation which can reduce cost, lead time, risk and CO_2 emission in a multimodal transportation system.

Column 1	2	3	4	5	6	7
	Local risk scores					
Segmented route	Freight damaged risks (0.56)	Infrastructure and equipment risks (0.26)	Operational risk (0.12)	Political and legislatorial risk (0.06)	Overall transportation risk score	Risk priority ranking
A_1	0.789475	0.564816	0.403512	0.333336	0.6573798	10
A_2	0.771931	0.722223	0.456143	0.382356	0.6977379	13
A_3	0.53509	0.490742	0.999999	0.980393	0.6060668	8
A_4	0.999999	0.712964	0.625042	0.696081	0.86214	18
A_5	0.947368	1.000000	0.701756	1.000001	0.9347369	19
A_6	0.684212	0.453704	0.57895	0.578434	0.6053018	7
A_7	0.877193	0.620373	0.835566	0.627455	0.7904403	17
A_8	0.842106	0.648149	0.807018	0.539219	0.7692934	16
A_9	0.771931	0.518522	0.456144	0.352944	0.643011	9
A_{10}	0.412283	0.453704	0.438599	0.627453	0.4391206	3
A_{11}	0.377195	0.287039	0.471613	0.480395	0.3712766	2
A_{12}	0.49123	0.361111	0.618502	0.509807	0.4737863	5
A_{13}	0.824562	0.555556	0.508774	0.656866	0.7066641	14
A_{14}	0.701756	0.851852	0.460609	0.450984	0.696797	12
A_{15}	0.614037	0.768519	0.736843	0.56863	0.6662146	11
A_{16}	0.999999	0.925926	0.859649	0.882355	0.9568394	20
A_{17}	0.877193	0.555556	0.771931	0.460788	0.7559516	15
A_{18}	0.403511	0.620371	0.947368	0.470591	0.5291822	6
A_{19}	0.403511	0.555557	0.524244	0.323532	0.4527322	4
A_{20}	0.333336	0.416668	0.436525	0.313728	0.3662085	1

Table XIII.
Local and overall risk scores of the 20 segmented routes

Route	Segmented route 1	2	3	4	5	6	Overall transportation risk scores of each route	Risk priority ranking
1. BKK###HMC—Da Nang	A_1	A_2	A_3	A_4	A_5	A_6	4.363363	9
2. BKK###HMC +++ Da Nang	A_1	A_2	A_3	A_7	A_8	A_9	4.163929	8
3. BKK***Da Nang	A_{10}	A_{11}	A_{12}				1.284184	1
4. BKK###Da Nang	A_1	A_{13}	A_{14}				2.060841	2
5. BKK—HMC###Da Nang	A_{15}	A_{16}	A_4	A_3	A_{17}	A_{14}	4.544009	10
6. BKK—HMC +++ Da Nang	A_{15}	A_{16}	A_4	A_7	A_8	A_9	4.687939	11
7. BKK—HMC***Da Nang	A_{15}	A_{16}	A_4	A_{18}	A_{19}	A_{12}	3.940895	7
8. BKK***HMC +++ Da Nang	A_{10}	A_{20}	A_{18}	A_7	A_8	A_9	3.537256	5
9. BKK***HMC—Da Nang	A_{10}	A_{20}	A_{18}	A_4	A_5	A_6	3.736690	6
10. BKK***HMC###Da Nang	A_{10}	A_{20}	A_{18}	A_3	A_{17}	A_{14}	3.393327	3
11. BKK###HMC***Da Nang	A_1	A_2	A_3	A_{18}	A_{19}	A_{12}	3.416885	4

Notes: —, Truck; +++, train; ***air and ###, sea

Table XIV.
The results of combination between links and nodes quantitative risk for each route

Therefore, there are four criteria which are integrated in the objective function of zero-one goal programming. The four criteria consist of transportation cost, transportation time, EI and transportation risk. The pairwise comparison matrix for the four criteria provided by the LSPs is as follows (Table XVI).

Route	Transportation cost (USD/20 ft.)	Transportation time (days)	Environmental impact scale	Transportation risk
1. BKK###HMC—Da Nang	2,967.5	8	2	4.363363
2. BKK###HMC +++ Da Nang	1,631.5	10	3	4.163929
3. BKK***Da Nang	4,580	2	1	1.284184
4. BKK###Da Nang	1,080	8	2	2.060841
5. BKK—HMC###Da Nang	4,298	7	2	4.544009
6. BKK—HMC +++ Da Nang	4,254	7	4	4.687939
7. BKK—HMC***Da Nang	7,880	4	2	3.940895
8. BKK***HMC +++ Da Nang	5,334	6	3	3.537256
9. BKK***HMC—Da Nang	6,580	4	2	3.736690
10. BKK***HMC###Da Nang	5,378	6	2	3.393327
11. BKK###HMC***Da Nang	5,257.5	7	2	3.416885

Table XV.
Database of multimodal transportation route

The relative weight criteria from AHP are of transportation cost 0.56, transportation time 0.23, EI 0.05 and transportation risk 0.16 and the maximum eigenvalue (λ_{max}) is 4.25. The consistency index for the above paired comparison matrix (*CI*) is 0.083 and the corresponding *CR* is 0.093. Because the corresponding consistency is <0.1 ($CR<0.1$), the pairwise comparison matrix is considered to have an acceptable consistency.

After defining the relative weight criteria, the significant weight of each criterion is integrated in the objective function of ZOGP.

3.6 Optimization by using ZOGP methodology

Finally, the ZOGP methodology can be optimized to obtain a multimodal transportation route. The problem is formulated by using Equations (13)-(18) in Section 2.6. The significant weight obtained via the AHP method in the previous phase is added into the objective function of ZOGP. The significant weight from AHP, parameters and limited data from previous phases are used to formulate the objective function and constraints.

In this case study, the limitations of criteria are set as constraints that consist of the budget not over 5,000 USD, lead time not over eight days, EI index at least 3, transportation risk score not over 3.5 and the relative weight criteria from AHP method as budget 0.56, transportation time 0.23, EI 0.05 and transportation risk 0.16 with CR not over 0.1 to find the optimal route.

Because all the data in each objective function of this research has a different unit, all units are converted into percentage. More details of the converting of all units in percentage can be seen in Appendix 3. The model of integrated AHP and ZOGP for calculating an optimal multimodal transportation route is presented below:

$$\text{Minimize} = \sum_{i=1}^{m} \left(g_i d_i^+\right) = 0.56\left(d_1^+\right) + 0.23\left(d_2^+\right) + 0.05\left(d_3^+\right) + 0.16\left(d_4^+\right) \tag{19}$$

$$\text{Subject to: Budget} \quad \begin{aligned} &40.65X_1 + 67.37X_2 + 8.4X_3 + 78.4X_4 + 14.04X_5 + 14.92X_6 \\ &- 57.6X_7 - 6.68X_8 - 31.6X_9 - 7.56X_{10} - 5.15X_{11} - d_1^+ = 78.4 \end{aligned} \tag{20}$$

$$\text{Time} \quad \begin{aligned} &100X_1 + 125X_2 + 25X_3 + 100X_4 + 87.5X_5 + 87.5X_6 \\ &+ 50X_7 + 75X_8 + 50X_9 + 75X_{10} + 87.5X_{11} - d_2^+ = 100 \end{aligned} \tag{21}$$

Table XVI.
The pairwise comparison matrix for the four criteria provided by the LSPs

	Transportation cost	*Transportation time*	*Environmental impact*	*Transportation risk*
$A =$				
Transportation Cost	1	3	7	5
Transportation Time	1/3	1	5	2
Environmental Impact	1/7	1/5	1	1/5
Transportation Risk	1/5	1/2	5	1
Consistency Matrix $A =$				
Transportation Cost	1	3	7	5
Transportation Time	1/3	1	5	2
Environmental Impact	1/7	1/5	1	1/5
Transportation Risk	1/5	1/2	5	1
Summation in Column	1.68	4.7	18	8.2
Mean Matrix $A =$				
Transportation Cost	$1/1.68 = 0.60$	$3/4.7 = 0.64$	$7/18 = 0.39$	$5/8.2 = 0.61$
Transportation Time	$\frac{1}{3}/1.68 = 0.20$	$1/4.7 = 0.21$	$5/18 = 0.28$	$2/8.2 = 0.24$
Environmental Impact	$\frac{1}{7}/1.68 = 0.09$	$\frac{1}{5}/4.7 = 0.04$	$1/18 = 0.06$	$\frac{1}{5}/8.2 = 0.02$
Transportation Risk	$\frac{1}{5}/1.68 = 0.12$	$\frac{1}{2}/4.7 = 0.11$	$5/18 = 0.28$	$1/8.2 = 0.12$
Weight Criteria $=$				
Transportation Cost	$= (0.60 + 0.64 + 0.39 + 0.61)/4$	$= 0.56$		
Transportation Time	$= (0.20 + 0.21 + 0.28 + 0.24)/4$	$= 0.23$		
Environmental Impact	$= (0.09 + 0.04 + 0.06 + 0.02)/4$	$= 0.05$		
Transportation Risk	$= (0.12 + 0.11 + 0.28 + 0.12)/4$	$= 0.16$		

$$\text{Environmental Impact } 33.33X_1 + 66.67X_3 + 33.33X_4 + 33.33X_5 - 33.33X_6 + 33.33X_7 + 33.33X_9 + 33.33X_{10} + 33.33X_{11} - d_3^+ = 66.67 \quad (22)$$

$$\text{Risk} - 24.67X_1 - 18.97X_2 + 63.31X_3 + 41.12X_4 - 29.83X_5 - 33.94X_6 - 12.60X_7 - 1.06X_8 - 6.76X_9 + 3.05X_{10} + 2.37X_{11} - d_4^+ = 63.31 \quad (23)$$

$$\begin{aligned} &X_1 + X_2 + \cdots + X_n = 1 \\ &w_i d_i^+ \geqslant 0, \ \textit{for } i = 1, 2, \ldots, m \\ &c_j, t_j, e_j r_j \geqslant 0, \ \textit{for } j = 1, 2, \ldots, n \\ &X_j = 0 \textit{ or } 1, \ \textit{for } j = 1, 2, \ldots, n \end{aligned} \quad (24)$$

In this research, there are four goals. The first goal is the transport budget (g_1) that is not over the user limit. The second goal is the transport time (g_2) that is not over the lead time limited by the user. The third goal is the EI (g_3) that is less than the minimum scale which is limited by the user. Finally, the fourth goal is the multimodal transportation risk (g_4) that is less than the user limit.

The deviation variables (d_i^+) in Equation (19) are the overachievement percentage vectors of each goal. w_i are the significant weights for each criteria coefficient relating to deviation of each goal that can obtain from AHP ($w_1 = 0.56$ is weight of budget criterion, $w_2 = 0.23$ is weight of time criterion, $w_3 = 0.05$ is weight of EI criterion and $w_4 = 0.16$ is weight of risk criterion).

X_j represents the zero-one variables representing the non-selection (i.e. a zero) or selection (i.e. a one) of route $j = 1, 2, 3, \ldots, n$, subject to criteria constraint in the right hand side (budget, time, EI and risk).

In Equation (20)-(23) are the constraint equations of the model. Finally, Equation (24) can be used to control the result that only one route is optimum for one situation (Kengpol *et al.*, 2012a).

This mathematic model of integrated AHP and ZOGP is solved on a spreadsheet software. The results found that the optimal route is sea transport mode departing from Bangkok to Da Nang Port (Route 4). Transportation cost is equal to 1,080 USD for an 8-day period of transportation, an EI scale is equal to 2 and transportation risk score is equal to 2.06. If the user needs to find the next optimal route, the user can recalculate by deleting the prior optimal route which has been illustrated in Table XVII.

Route	Transportation cost (USD/20 ft.)	Transportation time (days)	Environmental impact scale	Transportation risk
1. BKK###Da Nang (Route 4)	1,080	8	2	2.060841
2. BKK***Da Nang (Route 3)	4,580	2	1	1.284184

Note: ***Air and ###, sea

Table XVII. The result of the optimal route priority ranking

4. Conclusions, recommendations and future study

The objective of this research is to formulate a mathematical model to optimize a multimodal transportation route which can reduce cost, lead time, CO_2 emission and transportation risk in a multimodal transportation system. In order to achieve that, the development of a framework for route selection in multimodal transportation from both LSPs and customers is proposed. This new conceptual framework for route selection in multimodal transportation consists of six main phases. In this research, the multimodal transport cost model is used to calculate transportation cost and transportation time. The CO_2 emissions are calculated based upon the 2006 guidelines of IPCC. The risk calculation begins with the LSPs analyzing a nature of multimodal transportation risk by using FMEA methodology. Then, the quantitative risk analysis and risk picture diagram are used to assess the risk level and define a set of assessment grades in linguistic terms. Next, a new integrated AHP-DEA methodology is used to assess the risks of a large number of alternatives. This proposed methodology applies the AHP to determine the weights of the criteria and linguistic terms of assessment grades. In addition, this research also applies the SAW method to aggregate local risk scores into an overall transportation risk score for each decision alternative. After that, the AHP method is used again to prioritize criteria. The significant weight of criteria obtained from AHP can be integrated in the objective function of ZOGP. Finally, the zero-one goal programming is used to generate the optimal route. The new conceptual framework for route selection in multimodal transportation has been tested on a realistic multimodal transportation service, originating from Bangkok in Thailand to a destination at Da Nang port in Vietnam. The results found that the optimal route is sea transport mode departing from Bangkok to Da Nang Port (Route 4). Transportation cost is equal to 1,080 USD for an 8-day period of transportation, an EI scale is equal to 2 and transportation risk score is equal to 2.06.

The contribution of this research lies in the development of a new decision support approach that is flexible and applicable to LSPs, in selecting a multimodal transportation route under the multi-criteria in terms of cost, time, risk and importantly the EI. The result shows that the new approach can provide guidance in choosing the lowest cost route in accordance with other criteria, and to minimize the CO_2 emission effectively. The advantage of this research is that the user can select the optimal multimodal transportation route and set the significant weight as needed. Furthermore, in the risk calculation, the framework has an advantage as the user can reduce the bias of risk assessment on each multimodal transportation route because the new conceptual framework utilizes historical data to identify rank of probability assessment scale and the severity impact assessment scale.

For further study, this research plans to develop a new algorithm to solve the multimodal transport problem when the problem has a larger scale of alternatives.

References

Afshari, A., Mojahed, M. and Yusuff, R.M. (2010), "Simple additive weighting approach to personnel selection problem", *International Journal of Innovation, Management and Technology*, Vol. 1 No. 5, pp. 511-515.

Ayed, H., Fernandez, C.G., Habbas, Z. and Khadraoui, D. (2011), "Solving time-dependent multimodal transport problems using a transfer graph model", *Computer & Industrial Engineering*, Vol. 61 No. 2, pp. 391-401.

Banomyong, R. and Beresford, A.K.C. (2001), "Multimodal transport: the case of Laotian garment exporters", *International Journal of Physical Distribution & Logistics Management*, Vol. 31 No. 9, pp. 663-685.

Bielli, M., Boulmakoul, A. and Mouncif, H. (2006), "Object modeling and path computation for multimodal travel systems", *European Journal of Operational Research*, Vol. 175 No. 3, pp. 1705-1730.

Chang, T.S. (2008), "Best routes selection in international intermodal network", *Computers & Operations Research*, Vol. 35 No. 9, pp. 2877-2891.

Charnes, A., Cooper, W.W. and Rhodes, E. (1978), "Measuring the efficiency of decision making units", *European Journal of Operational Research*, Vol. 2 No. 6, pp. 429-444.

Eggleston, H.S., Buendia, L., Miwa, K., Ngara, T. and Tanabe, K. (2006), "2006 IPCC guidelines for national greenhouse gas inventories", Intergovernmental Panel on Climate Change National Greenhouse Gas Inventories Programme, Institute for Global Environmental Strategies.

Hallikas, J., Karvonen, I., Pulkkinen, U., Virolainen, V.M. and Tuominen, M. (2004), "Risk management processes in supplier networks", *International Journal of Production Economics*, Vol. 90 No. 1, pp. 47-58.

Ham, H., Kim, T.J. and Boyce, D. (2005), "Implementation and estimation of a combined model of interregional, multimodal commodity shipments and transportation network flows", *Transportation Research Part B*, Vol. 39 No. 1, pp. 65-79.

Hao, H., Wang, H. and Ouyang, M. (2012), "Fuel consumption and life cycle GHG emissions by China's on-road trucks: future trends through 2050 and evaluation of mitigation measures", *Energy Policy*, Vol. 43, pp. 244-251.

International Energy Agency (2009), *World Energy Outlook*, IEA, Paris.

Janic, M. (2007), "Modelling the full costs of an intermodal and road freight transport network", *Transportation Research Part D*, Vol. 12 No. 1, pp. 33-44.

Jirajariyawet, A. (2010), *Carbon Footprint of Products (CFP)*, National Metal and Materials and Technology Center (MTEC), Thai.

Kengpol, A., Meethom, W. and Tuominen, M. (2012a), "The development of a decision support system in multimodal transportation routing within Greater Mekong sub-region countries", *International Journal of Production Economics*, Vol. 140 No. 2, pp. 691-701.

Kengpol, A., Tuammee, S. and Tuominen, M. (2012b), "Analysis and design of management system for multimodal transportation", *Proceedings of the Asia Pacific Industrial Engineering & Management Systems Conference, Phuket, December 2-5*, pp. 136-144.

Liu, F.H.F. and Hai, H. (2005), "The voting analytic hierarchy process method for selecting supplier", *International Journal of Production Economics*, Vol. 97 No. 3, pp. 308-317.

Liu, H.C., Liu, L. and Liu, N. (2013), "Risk evaluation approaches in failure mode and effects analysis: a literature review", *Expert Systems with Applications*, Vol. 40 No. 2, pp. 828-838.

Min, H. (1991), "International intermodal choices via chance-constrained goal programming", *Transportation Research Part A*, Vol. 25 No. 6, pp. 351-362.

Modarres, M., Kaminskiy, M. and Krivtsov, V. (1999), *Reliability Engineering and Risk Analysis*, Marcel Dekker Inc., New York, NY.

Mohajeri, N. and Amin, G.R. (2010), "Railway station site selection using analytical hierarchy process and data envelopment analysis", *Computers & Industrial Engineering*, Vol. 59 No. 1, pp. 107-114.

Noguchi, H., Ogawa, M. and Ishii, H. (2002), "The appropriate total ranking method using DEA for multiple categorized purposes", *Journal of Computational and Applied Mathematics*, Vol. 146 No. 1, pp. 155-166.

Qu, L. and Chen, Y. (2008), "A hybrid MCDM method for route selection of multimodal transportation network", *Proceedings of the 5th International Symposium on Neural Networks: Advances in Neural Networks, Beijing, September 24-28*, Springer-Verlag, Berlin and Heidelberg, pp. 374-383.

Ramanathan, R. (2001), "Comparative risk assessment of energy supply technologies: a data envelopment analysis approach", *Energy*, Vol. 26 No. 2, pp. 197-203.

Ramanathan, R. (2006), "Data envelopment analysis for weight derivation and aggregation in the analytic hierarchy process", *Computer & Operations Research*, Vol. 33 No. 5, pp. 1289-1307.

Reniers, G.L.L. and Dullaert, W. (2013), "A method to assess multi-modal hazmat transport security vulnerabilities: hazmat transport SVA", *Transport Policy*, Vol. 28, pp. 103-113.

Saaty, T.L. (1990), *Decision Making: The Analytic Hierarchy Process*, RWS Publications, Pittsburgh, PA.

Scenna, N.J. and Cruz, A.S.M.S. (2005), "Road risk analysis due to the transportation of chlorine in Rosario city", *Reliability Engineering and System Safety*, Vol. 90 No. 1, pp. 83-90.

Soltanifar, M. and Lotfi, F.H. (2011), "The voting analytic hierarchy process method for discriminating among efficient decision making units in data envelopment analysis", *Computers & Industrial Engineering*, Vol. 60 No. 4, pp. 585-592.

Soons, C.J., Bosch, J.W., Arends, G. and Gelder, P.H.A.J.M.V. (2006), "Framework of a quantitative risk analysis for the fire safety in metro systems", *Tunnelling and Underground Space Technology*, Vol. 21 Nos 3-4, pp. 281-287.

Southworth, F. and Peterson, B.E. (2000), "Intermodal and international freight network modeling", *Transportation Research Part C*, Vol. 8 Nos 1-6, pp. 147-166.

The UNCTAD secretariat (2003), "Development of multimodal transport and logistics services", *Proceedings of United Nations Conference on Trade and Development, Geneva*, pp. 1-20.

Tillman, F.A. and Cassone, D.T. (2012), *A Professional's Guide to Decision Science and Problem Solving*, FT Press, New Jersey, NJ.

Timilsina, G.R. and Shrestha, A. (2009), "Transport sector CO_2 emissions growth in Asia: underlying factors and policy options", *Energy Policy*, Vol. 37 No. 11, pp. 4523-4539.

Tsai, M.C. and Su, C.C. (2004), "Scenario analysis of freight vehicle accident risks in Taiwan", *Accident Analysis and Prevention*, Vol. 36 No. 4, pp. 683-690.

Verma, M. (2011), "Railroad transportation of dangerous goods: a conditional exposure approach to minimize transport risk", *Transportation Research Part C*, Vol. 19 No. 5, pp. 790-802.

Vilko, J.P.P. and Hallikas, J.M. (2012), "Risk assessment in multimodal supply chains", *International Journal of Production Economics*, Vol. 140 No. 2, pp. 586-595.

Vose, D. (2000), *Risk Analysis: A Quantitative Guide*, John Wiley & Sons Ltd, Chichester.

Wang, Y.M., Liu, J. and Elhag, T.M.S. (2008), "An integrated AHP-DEA methodology for bridge risk assessment", *Computers & Industrial Engineering*, Vol. 54 No. 3, pp. 513-525.

Yabe, N. (2004), "An analysis of CO_2 emissions of Japanese industries during the period between 1985 and 1995", *Energy Policy*, Vol. 32 No. 5, pp. 595-610.

Zhang, X., He, Z. and Pan, Y. (2010), "Study on multimodal transport network model base on genetic algorithm method", *Proceedings of ICLEM*, pp. 3517-3523.

Further reading

Charnes, A., Cooper, W.W. and Ferguson, R.O. (1955), "Optimal estimation of executive compensation by linear programming", *Management Science*, Vol. 1 No. 2, pp. 138-151.

Donner, P. (2008), "Managing risks of multimodal transport operators", *Proceedings of 1st Arab Logistics & Multimodal Transport Conference, Amman*, pp. 1-8.

Giglio, D., Minciardi, R., Pizzorni, D., Rudari, R., Sacile, R., Tomasoni, A. and Trasforini, E. (2004), "Towards a decision support system for real time risk assessment of hazardous material transport on road", *Proceedings of the International Environmental and Monitoring Software Society Conference, University of Osnabrück, Germany, June 14-17*, pp. 1-6.

Hendershot, D.C. (1999), "A team approach to transportation risk management", *Proceedings for Presentation at the Managing Transportation Risk as a Core Business Practice Conference, Baltimore*, pp. 1-9.

Tabucanon, M.T. (1988), *Multiple Criteria Decision Making in Industry*, Elsevier Science Publishers B.V, Amsterdam.

Taha, H.A. (2003), *Operations Research an Introduction*, Pearson Education Inc., Singapore.

Thailand Greenhouse Gas Management Organization (Public Organization) (2010), *The National Guideline Carbon Footprint of Product*, MTEC, Bangkok.

Wang, Y.M., Chin, K.S. and Yang, J.B. (2007), "Three new models for preference voting and aggregation", *Journal of the Operational Research Society*, Vol. 58 No. 10, pp. 1389-1393.

Appendix 1. CO_2 emission calculation

The CO_2 emission coefficient of each transport type (Emission Factor) has been illustrated in Table AI.

Transport type	Unit	Emission factor (kg CO_2 e/unit)
Truck		
10-Wheel container truck – 16 ton (no load)	km	0.6001
10-Wheel container truck – 16 ton (50% load)	ton-km	0.0887
10-Wheel container truck – 16 ton (75% load)	ton-km	0.0614
10-Wheel container truck – 16 ton (full load)	ton-km	0.0473
Ship		
Bulk	ton-km	0.002
Container	ton-km	0.0100
Train		
Train	ton-km	0.1111
Aircraft		
Short flight (0-400 Km)	ton-km	0.0002
Medium flight (401-1,300 Km)	ton-km	0.0002
Long flight (1,301-4,500 Km)	ton-km	0.0001
Extended flight (more than 4,501 Km)	ton-km	0.0001

Source: Jirajariyawet (2010)

Table AI. The CO_2 emission coefficient of each transport type

Appendix 2. Failure mode and effect analysis (FMEA)

Probability of failure	Possible failure rates	Rank
Extremely high	$\geqslant$ in 2	10
Very high	1 in 3	9
Repeated failures	1 in 8	8
High	1 in 20	7
Moderately high	1 in 80	6
Moderate	1 in 400	5
Relatively low	1 in 2,000	4
Low	1 in 15,000	3
Remote	1 in 150,000	2
Nearly impossible	$\leqslant$1 in 1,500,000	1

Source: Liu *et al.* (2013)

Table AII.
Suggested ratings for the occurrence of a failure mode

Effect	Criteria: severity of effect	Rank
Hazardous	Safety related failure modes causing noncompliance with government regulations without warning	10
Serious	Safety related failure modes causing noncompliance with government regulations with warning	9
Extreme	Failure modes resulting in loss of primary vehicle/system/component function	8
Major	Failure modes resulting in a reduced level of vehicle/system/component performance and customer dissatisfaction	7
Significant	Failure modes resulting in loss of function by comfort/convenience systems/components	6
Moderate	Failure modes resulting in a reduced level of performance of comfort/convenience systems/components	5
Low	Failure modes resulting in loss of fit and finish, squeak and rattle function	4
Minor	Failure modes resulting in partial loss of fit and finish, squeak and rattle functions	3
Very minor	Failure modes resulting in minor loss of fit and finish, squeak and rattle functions	2
None	No effect	1

Source: Modarres *et al.* (1999)

Table AIII.
Suggested ratings for the severity of a failure mode

Detection	Criteria: likelihood of detection by design control	Rank
Absolute uncertainty	Design control does not detect a potential cause of failure mode; or there is not design control	10
Very remote	Very remote chance the design control will detect a potential cause of failure mode	9
Remote	Remote chance the design control will detect a potential cause of failure mode	8
Very low	Very low chance the design control will detect a potential cause of failure mode	7
Low	Low chance the design control will detect a potential cause of failure mode	6
Moderate	Moderate chance the design control will detect a potential cause of failure mode	5
Moderately high	Moderately high chance the design control will detect a potential cause of failure mode	4
High	High remote chance the design control will detect a potential cause of failure mode	3
Very high	Very high chance the design control will detect a potential cause of failure mode	2
Almost certain	Design control will almost certainly detect a potential cause of failure mode	1

Source: Liu *et al.* (2013)

Table AIV.
Suggested ratings for the detection of a failure mode

Appendix 3. Example for calculation of the converted values of all units in percentage

In order to learn how to "convert all units in percentage," this section presents the example for calculation of the converted values of all units in percentage as below.

This case study sets the constraints as the budget not over 5,000 USD, lead time not over eight days, environmental impact index at least 3 and transportation risk score not over 3.5.

For example, according to Equation (20), the coefficient of X_j in budget constraint (c_j) is a cost of each route in percentage of the user budget which can be calculated as follows (Cost of route see Table III):

$$c_j = \left[\frac{(\text{Budget limited by user}) - (\text{Cost of route } j)}{(\text{Budget limited by user})}\right] \times 100$$

$$c_1 = \left[\frac{(5{,}000) - (2{,}967.5)}{(5{,}000)}\right] \times 100 = 40.65$$

$$c_2 = \left[\frac{(5{,}000) - (1{,}631.5)}{(5{,}000)}\right] \times 100 = 67.37$$

$$c_3 = \left[\frac{(5{,}000) - (4{,}580)}{(5{,}000)}\right] \times 100 = 8.4$$

$$c_4 = \left[\frac{(5{,}000) - (1{,}080)}{(5{,}000)}\right] \times 100 = 78.4$$

$$c_5 = \left[\frac{(5{,}000) - (4{,}298)}{(5{,}000)}\right] \times 100 = 14.04$$

$$c_6 = \left[\frac{(5{,}000) - (4{,}254)}{(5{,}000)}\right] \times 100 = 14.92$$

$$c_7 = \left[\frac{(5{,}000) - (7{,}880)}{(5{,}000)}\right] \times 100 = -57.6$$

$$c_8 = \left[\frac{(5{,}000) - (5,334)}{(5{,}000)}\right] \times 100 = -6.68$$

$$c_9 = \left[\frac{(5{,}000) - (6{,}580)}{(5{,}000)}\right] \times 100 = -31.6$$

$$c_{10} = \left[\frac{(5{,}000) - (5{,}378)}{(5{,}000)}\right] \times 100 = -7.56$$

$$c_{11} = \left[\frac{(5{,}000) - (5{,}257.5)}{(5{,}000)}\right] \times 100 = -5.15$$

The right hand side of Equation (20) is the percentage of the budget limited by the user which can be calculated as follows (for cost of route see Table III):

$$C = \left[\frac{(\text{Budget limited by user}) - (\text{Minimum cost of all route})}{(\text{Budget limited by user})}\right] \times 100$$

$$C = \left[\frac{(5{,}000) - (1{,}080)}{(5{,}000)}\right] \times 100 = 78.4$$

Therefore, the budget constraint equations of the model can be presented below.

$$\text{Budget: } 40.65X_1 + 67.37X_2 + 8.4X_3 + 78.4X_4 + 14.04X_5 + 14.92X_6 - 57.6X_7 - 6.68X_8 - 31.6X_9 - 7.56X_{10} - 5.15X_{11} - d_1^+ = 78.4$$

Similarly, the coefficient of X_j and the right hand side of other constraint can be computed by using equation in Section 2.6 (Parameters).

About the authors

Athakorn Kengpol is an Associate Professor of Industrial Engineering at the Faculty of Engineering, King Mongkut's University of Technology North Bangkok, Thailand. He received PhD in Manufacturing Engineering and Operations Management from The University of Nottingham, UK and DSc in Industrial Engineering and Management from the Lappeenranta University of Technology, Finland. His research interests include decision science, concurrent engineering, management information systems and packaging design. Associate Professor Athakorn Kengpol is the corresponding author and can be contacted at: athakorn@kmutnb.ac.th

Sopida Tuammee is a PhD Student in Department of Industrial Engineering, Faculty of Engineering, King Mongkut's University of Technology North Bangkok, Thailand. Her research interests include multimodal transportation and multiple criteria decision making (MCDM).

Markku Tuominen is a Professor of Industrial Management at the Faculty of Industrial Engineering and Management, Lappeenranta University of Technology, Finland. His research interests include decision support systems and methods of technology management.

To purchase reprints of this article please e-mail: **reprints@emeraldinsight.com**
Or visit our web site for further details: **www.emeraldinsight.com/reprints**